Ajanta Saha

PhD Candidate, Dept. of ECE, Purdue University, 465 Northwestern Ave, West Lafayette, Indiana -47907
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Objective

Seeking full-time engineering/research position in electrical/electrochemical device/circuit modeling, characterization, and reliability analysis.

Education

Purdue University, West Lafayette, Indiana, USA Jan 2020 – Apr 2024
PhD in Electrical and Computer Engineering | CGPA: 4.0/4.0

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh Oct 2017 – Dec 2019
Master of Science (M.Sc.) in Electrical and Electronic Engineering | CGPA: 3.92/4.0

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh Feb 2013 – Sep 2017
Bachelor of Science (B.Sc.) in Electrical and Electronic Engineering | CGPA: 3.98/4.0 (**1st among 214 students**)

Technical Skills

Programming Languages: C, C++, Python, Verilog
Circuit Simulator & Design Tools: Cadence (Virtuoso), PSPICE, Quartus, Proteus, MATLAB Simulink
Device Simulator & Numerical Analysis: COMSOL, MATLAB, Lumerical
Laboratory Characterization Tools: Potentiometer, EIS (Gamry Instruments), Oscilloscope, Microcontroller
Data Analytics: Statistical Analysis, Time Series Analysis, Machine Learning

Research Experience

Laboratory for Classical and Emerging Electronic Devices (CEED), Purdue University Jan 2020 – Present
Graduate Research Assistant | Advisor: Muhammad Ashraful Alam

Circuit design and characterization of dual sensor & energy harvester | skills: *EIS, PSPICE, Arduino, accelerated testing*

- Created energy harvester with potentiometric sensor and developed its circuit model using Electrochemical Impedance Spectroscopy (EIS) to project power yield, voltage stability, degradation, lifetime and aging effect
- Designed a microcontroller (Arduino) based accelerated testing system to experimentally validate circuit model
- Improved power yield by 400× by improving material characteristic of harvester guided by circuit model [5]

Physics-based analytical and numerical modeling of electrochemical sensor | skills: *COMSOL*

- Proposed analytical model to predict response time, sensor-to-sensor variation, signal-to-noise ratio of electrochemical sensors for healthcare/environmental applications by solving drift-diffusion (ODE/PDE) equations [6]
- Built coupled electro-chemical simulation framework in COMSOL to validate proposed model and experiential results
- Collaborated in developing physics (analytical model) guided ML framework for accelerated testing of sensor [9]

Time-series data analysis algorithm for high precision sensing | skills: *statistical methods, Python, pandas*

- Developed maximum likelihood estimation (MLE) based algorithm in Python to obtain actual signal from noisy data and measure credibility/reliability of sensors in continuous healthcare/environmental monitoring systems [1]
- Achieved up to 99% pH sensing accuracy by correcting radiation/temperature/bio-fouling induced drifts and reduced energy cost of edge IoT sensors by 3x using credibility based data transmission
- Improved accuracy of machine learning (ML) models for predicting sensor performance from image analysis by credibility based data filtering [8]

Physics-guided drift monitoring and calibration of sensor | skills: *modeling, experiments, reliability characterization*

- Invented thermometer-free analyte temperature estimation method only from voltage measurement of sensors [2-3]
- Characterized drift and degradation of electrochemical sensors under long-term thermal cycle
- Developed a physics-based drift detection and self-calibration algorithm using ambient temperature supervision to obtain nitrate concentration within 10% of laboratory precision in agricultural field [4][7]

Solid-state Electronics and Photonics (SSEP) research group, BUET Oct 2017 – Dec 2019
Graduate student | Advisor: Md Zunaid Baten

Indoor photovoltaic cell as energy harvester for IoT devices | skills: *MATLAB, Lumerical*

- Designed spectra-dependent photonic heterostructure with Finite-Difference Time-Domain (FDTD) in Lumerical to optimize performance of indoor photovoltaic cell under different light sources [10-11]
- Built coupled optical-electrical simulation framework and obtained 30% light absorption enhancement and 2% energy harvesting efficiency gain from photonic structure

Projects

Industrial production monitoring with power signal processing | skills: *Python, pandas* Jan 2022 –Present

- Developed a time series data analysis algorithm combining pattern matching and signal processing techniques (FFT, up sampling) for monitoring operation and health of industrial (part) manufacturing machines
- Obtained 98% part classification accuracy by processing 3 months of power consumption data and validated efficacy of algorithm in part counting and anomalous part detection tasks

8T-SRAM design with Computing-in-Memory (CiM) | skills: *Cadence(Virtuoso), schematic, layout* Aug 2023

- Implemented schematic and layout of 64×64 memory array with 8T bit cells and peripheral circuits to perform parallel eight 8-bit in-memory addition
- Performed physical verification and post layout simulation and achieved time latency of 1 ns and energy consumption of 351 fJ with 90% array area efficiency [12]

Hardware implementation of frequency up-converter | skills: *MATLAB Simulink, Proteus, C, Arduino* Sep 2017

- Designed and implemented circuit of a single-stage low (50 Hz) to high (1000 Hz) frequency AC-AC converter for solid state transformer in smart power grid

Relevant Coursework

- Semiconductor Materials and Heterostructures
- Reliability Physics of Nanoelectronic Transistors
- Advanced VLSI Devices
- MOS VLSI Design
- Analog Circuit Design
- Power Electronics
- Introduction to Mathematical Statistics
- Computational Models and Methods

Awards

Undergraduate University Gold Medal: Awarded for attaining highest CGPA in entire batch
Undergraduate Program Excellency Award : Awarded to top student of each department

Teaching Experience

Bangladesh University of Engineering and Technology (BUET), Bangladesh Sep 2017 – Jan 2020
Faculty Member, Dept. of Electrical and Electronic Engineering

- Taught electrical circuits, digital, and power electronics theory courses to undergraduate classes of 65 students each
- Supervised over 200 students in electronics and power electronics laboratory courses and projects in learning general lab practices, measurement methods, and design of circuits relevant to respective fields
- Complied coding assignments and guided 100 students to solve problems with numerical techniques in MATLAB

Selected Publications

[1] **A. Saha**, S. Sedaghat, S. Gopalakrishnan, J. Waimin, A. Yermembetova, N. Glassmaker, C. Mousoulis, A. Shakouri, A. Wei, R. Rahimi, M. A. Alam, A New Paradigm of Reliable Sensing with Field-Deployed Electrochemical Sensors Integrating Data Redundancy and Source Credibility, **Scientific Reports**, 2023.

[2] **A. Saha**, A. Yermembetova, Y. Mi, S. Gopalakrishnan, S. Sedaghat, J. Waimin, P. Wang, N. Glassmaker, C. Mousoulis, N. Raghunathan, S. Bagchi, R. Rahimi, A. Shakouri, A. Wei, M. A. Alam,Temperature Self-Calibration of Always-On, Field-Deployed Ion-Selective Electrodes Based on Differential Voltage Measurement, **ACS Sensors**, 2022.

[3] M. A. Alam, **A. Saha**, System and Method for Self-calibrating of Ion Selective Electrodes Based on Differential Voltage Measurement. (Applied for U.S. Patent)

[4] **A. Saha**, Y. Mi, N. Glassmaker, A. Shakouri, M. A. Alam, In Situ Drift Monitoring and Calibration of Field-Deployed Potentiometric Sensors Using Temperature Supervision, **ACS Sensors**, 2023.

[5] **A. Saha**, M. A. Alam, Signal as an Energy Source: Potentiometric Sensors as an Energy Harvester (Under review)

[6] X. Jin, **A. Saha**, H. Jiang, M. R. Oduncu, Q. Yang, S. Sedaghat, K. Maize, J. P. Allebach, A. Shakouri, N. Glassmaker, A. Wei, R. Rahimi, M. A. Alam, Steady-State and Transient Performance of Ion-Sensitive Electrodes Suitable for Wearable and Implantable Electro-Chemical Sensing, **IEEE Transactions on Biomed. Eng.**, 2022.

[7] **A. Saha**, S. Gopalakrishnan, J. Waimin, S. Sedaghat, Y. Mi, N. Glassmaker, M. Cakmak, A. Shakouri, R. Rahimi, M. A. Alam, Embrace the Imperfection: How Intrinsic Variability of Roll-to-Roll Manufactured Environmental Sensors Enable Self-Calibrating, High-Precision Quorum Sensing, Proceedings of the **ASME**, 2022.

[8] X. Wang, **A. Saha**, Y. Mi, A. Shakouri, M. A. Alam, G. T. Chiu, J. P. Allebach, Thin-Film Nitrate Sensor Performance Prediction Based on Image Analysis and Credibility Data to Enable a Certify As Built Framework, Proceedings of the **ASME**, 2022.

[9] S. C. Mouli, S. Sedaghat, M. R. Oduncu, **A. Saha**, R. Rahimi, M. A. Alam, A. Wei, A. Shakouri, B. Ribeiro, Physics-Informed Machine Learning for Accelerated Testing of Roll-to-Roll Printed Sensors, Proceedings of the **ASME**, 2022.

[10] **A. Saha**, K. A. Haque, M. Z. Baten, Performance evaluation of single-junction indoor photovoltaic devices for different absorber bandgaps under spectrally varying white light-emitting diodes, **IEEE Journal of Photovoltaics**, 2019.

[11] **A. Saha**, E. Maria, M. Z. Baten, Spectra dependent photonic structure design for energy harvesting by indoor photovoltaic devices, **AIP Advances**, 2022.

[12] **A. Saha**, J. B. Jahangir, 8T-SRAM with computing-in-memory (CiM) of addition of two 8-bit words, 2023.